

PRODUCT OWNER CASE STUDY

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Investment Guardian

Pre-trade protection for CFDs, crypto & precious metals — built entirely on Revolut's own AI stack.

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The Gap Between Sherlock and the Trade

Revolut protects money leaving the app. Nothing yet protects the moment it's staked on a Revolut-native position.

30%

Cut in investment-scam card losses since Sherlock's 2024 AI feature

€5M

AGCM fine (part of €11.5M total) — unclear pre-trade disclosure

2025–26

ECB flags product governance outpacing controls

Sherlock evaluates every card payment in under 50ms and already cuts investment-scam losses by 30%. But there is no equivalent layer for the moment a customer opens a first CFD, crypto, or precious-metals position — exactly where the AGCM's €5M investment-disclosure finding and the ECB's governance review are pointing.

Sources: Revolut, Feb 2024 · AGCM press release PS12974, 2 Apr 2026 · Financial Times, 10 Jun 2026

Why Doesn't This Already Exist?

Revolut runs 9 FinCrime AI agents in production today. A gap this specific has real reasons behind it — not an oversight.



Friction Sits on Revenue

Every Guardian intervention sits on a monetised trading flow. Sherlock's warnings sit on an outbound payment that earns Revolut nothing — a materially easier internal case to make.



Consent Changes Liability

A scammed customer is a victim. A customer who knowingly opened a leveraged position and lost is, legally and commercially, a different category — and a different obligation for Revolut.



False Accusations Are a Real Risk

Wrongly calling a licensed broker a scam isn't a bad recommendation — it's a potentially defamatory claim about a regulated business. See Guardrails, Section 8



Part of This Already Exists

MiFID II's appropriateness test already covers part of this ground for complex instruments like CFDs. See next slide.

Why build it anyway — The regulatory signals here show the cost of inaction is rising faster than the conversion cost of acting. The 3–5% conversion cap in H1 is the mechanism that keeps that trade-off honest.

What Already Exists — And Where Guardian Goes Further

CFDs are legally “complex instruments.” MiFID II Article 25(3) already requires an appropriateness test before a retail customer can trade them — Revolut runs this today.[12]

EXISTING APPROPRIATENESS TEST	INVESTMENT GUARDIAN
Static — one assessment at onboarding	Per-transaction — evaluates this specific deposit
Knowledge-based — asks what the customer knows	Behavioural too — combines with PRAGMA signals on what they're doing
Scam-blind — tests leverage comprehension, not counterparty fraud	Scam-aware — classifies the archetype, names the pattern
Generic warning — same disclosure for everyone	Tiered response — disclosure → cooling-off → specialist escalation

Investment Guardian: One Layer, Four Capabilities

Configured — not re-coded — per instrument, market, and risk policy. Plugs directly into rails Revolut already runs.



1 · Pre-Trade Risk & Understanding Score

Adaptive 3–5 question check on leverage & loss understanding, combined with behavioural signals into a Low / Medium / High score.



2 · Scam Archetype Classifier

Classifies likely patterns — fake broker, signal-group, romance-investment, high-yield crypto — using PRAGMA embeddings.



3 · Archetype-Tailored Intervention

Response scales with risk: concise disclosure, cooling-off, strong warning, or live fraud-specialist escalation.

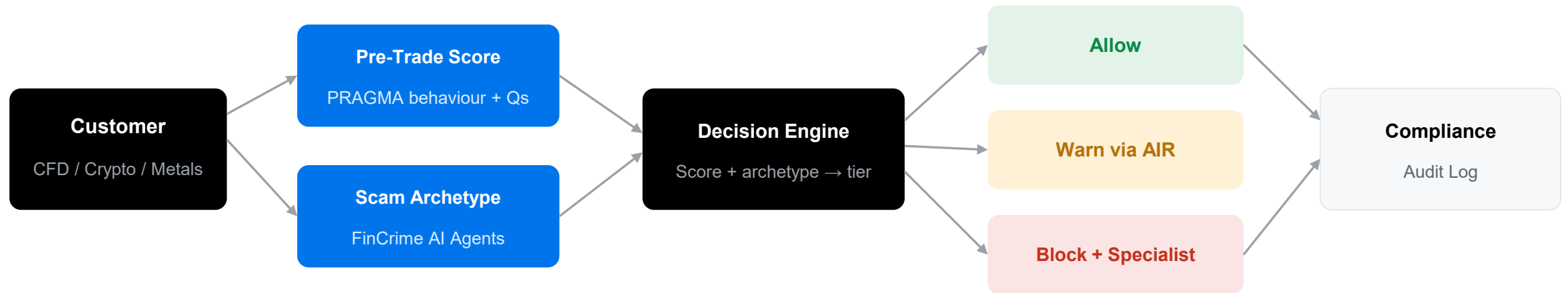


4 · Compliance-Ready Reporting

Structured logging extending existing FinCrime dashboards — audit-ready by default, no extra build.

How It Works

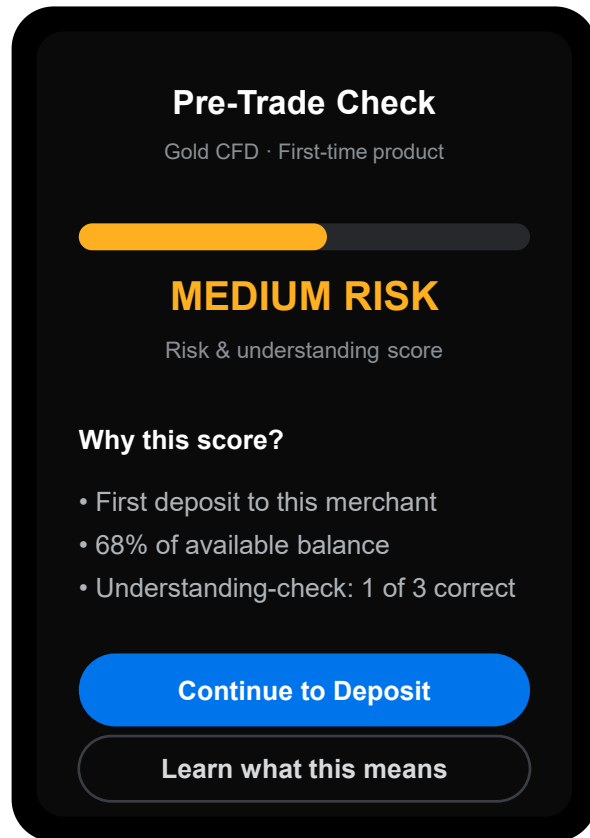
Every path — allow, warn, or block — terminates in the same immutable compliance log.



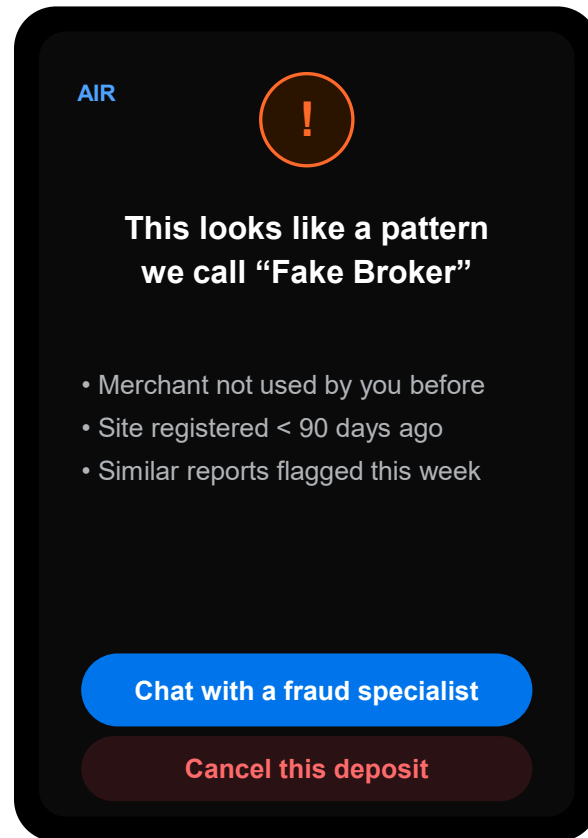
Platform, not patch — a configuration layer on PRAGMA, FinCrime AI Agents, and AIR. No new ML platform, no new case-management system.

How It Feels Inside the App

Marco funds a deposit to an unfamiliar Gold CFD broker. Guardian scores it Medium, flags “Fake Broker,” and AIR opens a specific conversation — not a generic disclaimer.



Screen 1 — Pre-Trade Risk Score



Screen 2 — AIR Intervention

Design principles

- Name the pattern, not just the risk
- Show the evidence behind the score
- Never fully block silently
- Respect the expert vs. novice split

Guardrails: When the Model Is Wrong

For a product whose output is an accusation, what happens when it's wrong is the central question — not a footnote.



Precision Target

≥90% precision required before the named archetype (e.g. “Fake Broker”) is shown. Below that: a generic, unnamed caution only. Illustrative target — to validate against real classifier performance.



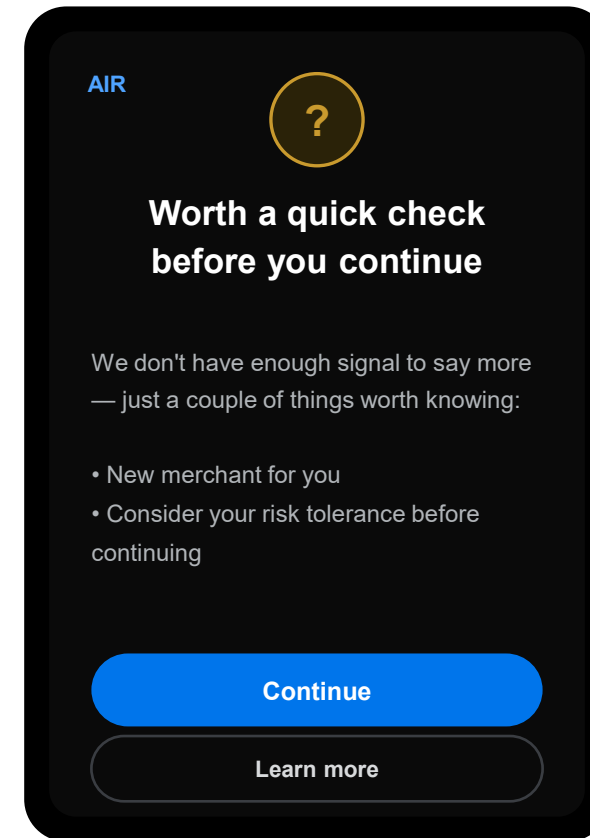
Confidence Tiering

High-confidence (≥90%) → names the pattern, as in Screen 2. Low-confidence → soft, generic warning with no named accusation — shown right.



Merchant Review Path

A flagged merchant can dispute the classification. Disputes route to a FinCrime ops queue with a 2-business-day SLA (illustrative) before the label is reinstated or cleared.



Screen 3 — Low-Confidence Generic Warning

Built on Revolut's Real Stack

Revolut invests in shared rails, not one-off systems. Guardian is specified against what these platforms do today.



PRAGMA

Revolut's transaction foundation model — transformer encoder, 10M–1B parameters, trained on 24bn events (207bn tokens) from 26M user histories, 111 countries.

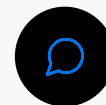
Guardian use: Behavioural half of the Risk Score.



FinCrime AI Agents

No-code platform to configure end-to-end AI agents with full observability. 9 agents already run AML/FinCrime decisions monthly.

Guardian use: Where the classifier is configured.



AIR

Revolut's in-app assistant, live to 13M UK customers since Apr 2026. Zero data retention; biometric-gated sensitive actions.

Guardian use: Delivers the Medium-tier warning.

Metrics Before Features

Every KPI carries an illustrative baseline; the guardrail that protects the business case is separated from the goals it's protecting.

Investment Scam Loss Rate

% of losses attributable to scams within high-risk flows

Illustrative baseline: ~0.9% of first deposits (placeholder — to replace with FinCrime's actual figure)

Complaint & Ombudsman Rate

Escalations per 10,000 high-risk investment transactions

Illustrative baseline: TBD — to source from Compliance

Override Rate & Trust Signal

% of Medium/High-warned users who proceed anyway (fast signal); segment NPS tracked as a slower cross-check

Illustrative baseline: TBD — override rate is the primary read; NPS lags

GUARDRAIL — not a goal. Conversion & Revenue: first-time deposit conversion for legitimate (Low-tier) users. Must not fall more than 3–5% vs. control (H1 cap). This protects the business case — it does not need to improve.

H1 — The pre-trade risk & understanding check reduces high-loss scam incidents on first-time CFD deposits by **≥20%**, with acceptable conversion impact capped at **3–5%**.

MVP = Capability 1 (Pre-Trade Score) only. Phase-2 hypotheses for Capabilities 2–4 → Section 9 (Day 1–90 Plan).

Operating From Day One

A concrete 90-day plan — discovery, scoped MVP, then algorithmisation — matching Revolut's own product-launch framework.

DAYS 1–30

Deep Discovery & Alignment

- Onboard into FinCrime AI Agents, PRAGMA, AIR
- Shadow fraud & investment teams
- Agree KPIs, hypotheses, cohorts

DAYS 31–60

MVP Build & Experiments

- MVP: Capability 1 only — Pre-Trade Score
- Instrument dashboards, launch A/B
- Iterate on quiz & intervention UX

DAYS 61–90

Algorithmisation & Scaling

- Present results in product review
- Abstract into configurable modules
- Agree scaling roadmap & governance

Phase 2 hypotheses (Capabilities 2–4)

H2 — Archetype Classifier — At $\geq 90\%$ precision, reduces confirmed-scam deposit completion by $\geq 15\%$ among flagged users.

H3 — Tailored Intervention — Named-pattern warnings produce a lower “proceed anyway” rate than generic warnings, at equivalent risk scores.

H4 — Compliance Reporting — Structured logging cuts manual case-prep time for FinCrime reporting (baseline TBD).

Why This Reflects How I'd Operate

Mapped to Revolut's actual values — Never Settle, expressed through four principles.



Think Deeper

Traced the problem to three dated, sourced regulatory signals — not a hunch.



Get It Done

Scoped to ship on existing rails within one quarter, not a platform rebuild.



Dream Team

A clear pod structure across Eng, Design, Data, Legal, and FinCrime.



Deliver Wow

Conversational, AIR-native interventions — not a static disclaimer wall.

Yash Todkari — Double Master's in International Management and Business Organization & Strategy, hands-on GenAI-for-product-discovery research, and an active CFD/metals trader — this case study was built from both sides of the modal.

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Let's talk.

I'd welcome the chance to walk through the archetype taxonomy, the experiment design, or the PRAGMA / FinCrime integration in more depth.

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Sources & Citations

Every Revolut-specific claim in this case study is sourced to a public press release, regulator filing, published paper, or news outlet — nothing here is internal or confidential Revolut data.

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